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Extracellular neural recordings from macaque primary and dorsal premotor motor cortex
during a sequential reaching task

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Summary

This data set includes extracellular recordings and behavior of macaque used in Lawlor, P.N. et al 2018 (under review as of 1/2018). The data was collected by Matthew G. Perich in the laboratory of Lee E. Miller at Northwestern University. The experiments examined the role of premotor cortex (PMd) and primary motor cortex (M1) in a sequential reaching task. In the experiment, the monkey controlled an on-screen cursor and was rewarded for moving that cursor to an indicated reach target, with multiple targets presented per trial. There were minimal kinematic requirements for the reaching movements (e.g., very brief hold times), meaning that the monkey typically made a relatively smooth series of reaches.

Behavioral variables recorded include position, velocity, and acceleration. Reach target locations and approximate presentation times are included. The electrophysiological recordings were collected with Utah multielectrode arrays from two monkeys (MM and MT). MM has data from one session, and MT has data from three sessions. There are tens of well-isolated units from both primary motor cortex (M1) and dorsal premotor cortex (PMd) for each session.

Contents:

- Behavior (instantaneous position, velocity, acceleration associated with reaches)
- Task information (reach target locations, approximate target onset timing)
- “Raw” format: Spike-sorted unit activity, otherwise unprocessed
- “Processed” format: Spike-sorted unit activity, more heavily processed and aligned to reach onset

The following paper analyzes a subset of the data from these experiments:

[1] Lawlor, P.N., Perich, M.G., Miller, L., Kording, K.P. Linear-Nonlinear-Time-Warp-Poisson models of neural activity. J Comput Neurosci (2018).
<https://doi.org/10.1007/s10827-018-0696-6>

Conditions for using the data

If you publish any work using the data, we request citation of [1] (referenced above) and also the data set using the following:

Matthew G. Perich, Patrick N. Lawlor, Konrad P. Kording, Lee E. Miller (2018).
Extracellular neural recordings from macaque primary and dorsal premotor motor cortex
during a sequential reaching task. . CRCNS.org.
<http://dx.doi.org/10.6080/K0FT8J72>

Methods

Adapted from [1]. See this paper for full methodological information.

Two monkeys (Monkey M, MM; Monkey T, MT) performed a reaching task in which they controlled a computer cursor using arm movements (Fig. 6). The monkey was seated in a primate chair while operating a two-link planar manipulandum. Hand movements were constrained to a horizontal plane within a workspace of 20 cm x 20 cm. In this task, an on-screen visual cue (2 cm square) specified the target location for each reach, and after making a series of four correct reaches to the targets, the monkey was given a liquid reward. The location of the target was pseudo-randomly¹ chosen to be located within an annulus (radius = 5-15 cm, angle = 360 degrees) centered on the current target.

The timing of target presentation was different for the first reach of each trial (beginning of series; made from rest) than for reaches two through four (mid-series). For the first reach, the target was presented and the monkey was allowed to move without an instructed delay period. For reaches two through four, the monkey initiated the next target by holding the cursor briefly within a 2 cm x 2 cm box centered on the current target. Upon reaching the current target, the next target was triggered (NB: not displayed) 100 ms later. The time of next target appearance was not precisely recorded, but was determined with later experiments to appear on average 96 ms after being triggered. Thus, on average, the next target appeared approximately 200 ms after the monkey reached the current target. Additionally, there was an imposed hold period of 100 ms that began at the same time as the triggering of the next target. Thus, there was a 200 ms time interval between when the monkey reached the current target and when it was allowed to initiate the next movement. This brief hold period had the effect of forcing the monkeys to decelerate as they approached the target. In practice, the task led to a series of relatively smooth arm movements of variable distance.

¹ Target locations were chosen as follows: 1) Randomly choose a distance between 5 and 15 cm, and an angle between 0° and 360° for the new target (relative to the current target). 2) If the new target falls outside of the workspace, add 90° to the angle and set the distance to be 5 cm. 3) Repeat step 2 until the target is in the workspace.

All surgical and experimental procedures were consistent with the guide for the care and use of laboratory animals and approved by the institutional animal care and use committee of Northwestern University. MM and MT were both implanted with 100-electrode arrays (Blackrock Microsystems, Salt Lake City, UT) in PMd (1mm electrode shaft length) and M1 (1.5mm length). Spike sorting was performed manually using standard techniques.

Data files organization

This data set includes data files at two levels of processing. The data format is described in more detail in the next section.

“Raw”: This is the least-heavily-processed data which includes sorted spikes, but is otherwise relatively raw. It includes all of the electrophysiological, behavioral, and task data recorded for each trial. There is one .mat file per recording session. Each of these .mat files includes three variables:

- M1: a structure containing the electrophysiological data and metadata for primary motor cortex (M1).
- PMd: a structure analogous to M1 but for premotor cortex (PMd)
- trial_table: a table containing information about each trial within the session.
- cont: a structure containing kinematic information for each trial

“Processed”: This is more heavily processed data. Rather than all recorded data, it includes only a window of electrophysiological and behavioral data associated with each reaching movement. Reaches have been extracted from each trial (there are multiple reaches per trial, up to four). This data file also includes various other quantities which were derived from the raw behavioral data (e.g, reach direction, reach starting and ending locations). The data are down-sampled. Some reaches have also been excluded for not having met certain quality requirements (these are described in the manuscript and code).

“Raw” files

File name	Monkey	# trials	# neurons, M1	# neurons, PMd
MM_S1_raw.mat	MM	140	67	94
MT_S1_raw.mat	MT	116	0	49
MT_S2_raw.mat	MT	178	0	46
MT_S3_raw.mat	MT	174	0	57

“Processed” files

File name	Monkey	# reaches	# neurons, M1	# neurons, PMd
MM_S1_processed.mat	MM	496	67	94
MT_S1_processed.mat	MT	419	0	49

MT_S2_processed.mat	MT	646	0	46
MT_S3_processed.mat	MT	652	0	57

Data format

“Raw” data (standard container (Matlab .mat file), custom organization)

- Data from each experimental session is contained in a single .mat file. Such a file contains all of the behavioral and electrophysiological data for each trial from that session. This file contains three Matlab variables. There are the electrophysiological data (in **M1** and **PMd**, depending on the particular experiment) and the experimental data (in **trial_table**).
 - **M1**: A Matlab structure with two sub-structures with information related to the M1 units.
 - **M1.units**: Contains the electrophysiological data for the units in M1 as well as some metadata. Each structure at this level (e.g., **M1.units(10)**) contains the data for that unit (e.g., unit 10).
 - **M1.units.id**: A 1 x 2 array; [*electrode_number*, *unit_ordinal*]. Gives the electrode number that the unit was recorded from as well as whether the unit was the first (1), second (2), etc. unit recorded from that electrode. Note: the unit ordinal is not the same as the unit number passed to the M1 struct.
 - **M1.units.wf**: A 48 x #spikes array. Gives the waveform for each spike of that unit.
 - **M1.units.ts**: A 1 x #spikes array. Gives the time of each spike.
 - **M1.units.misi**: Mean inter-spike interval (ISI) for this unit
 - **M1.units.offline_sorter_channel**: In the offline sorting software, the channel of this unit
 - **M1.sg**: A #units x 2 array; [*electrode_number*, *unit_ordinal*]. A redundant record of the unit ID for all units. Row number in this array is equal to the unit number in the **M1.units** structure.
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 - **PMd**: A Matlab structure with the same structure as **M1**. Contains information related to the PMd units.
 - **trial_table**: a Matlab array (#trials x 23) with the metadata for each trial. Unless otherwise noted, all times are recorded in seconds. The following information is contained in each column:
 - 1: trial start time (in seconds)
 - 2-6: information associated with target 1 (of up to 4 targets)

- 2: time of target1 appearance (uncorrected²)
- 3: time of movement onset towards target1
- 4: time of peak velocity towards target1
- 5: x location, target1 (arbitrary on-screen units)
- 6: y location, target1 (arbitrary on-screen units)
- 7-11: information associated with target 2
- 7: time of target2 appearance (uncorrected²)
- 8: time of movement onset towards target2
- 9: time of peak velocity towards target2
- 10: x location, target2 (arbitrary on-screen units)
- 11: y location, target2 (arbitrary on-screen units)
- 12-16: information associated with target 3
- 12: time of target3 appearance (uncorrected²)
- 13: time of movement onset towards target3
- 14: time of peak velocity towards target3
- 15: x location, target3 (arbitrary on-screen units)
- 16: y location, target3 (arbitrary on-screen units)
- 17-21: information associated with target4
- 17: time of target4 appearance (uncorrected²)
- 18: time of movement onset towards target4
- 19: time of peak velocity towards target4
- 20: x location, target4 (arbitrary on-screen units)
- 21: y location, target4 (arbitrary on-screen units)
- 22: time of trial end
- 23: trial result; 82 = Reward (successful trial), 65 = Abort (fails trial prior to the go cue, e.g., by leaving the center of work space during delay period), 70 = Fail (fails trial after the go cue, e.g., by not making it to the target in time), 74 = Incomplete (e.g., fails trial by not holding on target for enough time)
- **cont:** a Matlab structure containing timing and kinematic information about each trial. Original data was recorded at 1 kHz.
 - **cont.t:** array (#time_bins x 1) containing time stamps for each bin
 - **cont.pos:** array (#time_bins x 2) containing the x (column 1) and y (column 2) positions of the arm for each time bin

² The data acquisition system was not able to record the precise time of target presentation to the monkey. The system recorded the time of the command signal that the system sent to the display, but not the exact time the target was displayed on the screen. In practice, there is a delay between the command signal and actual appearance of target. This delay is also variable across trials, depending on factors such as the monitor refresh rate. Ideally, one would include, for example, a photodiode in the system that measures the appearance time of the target and records that. Our system lacked such a sensor at the time. Later experiments allowed us to estimate the average value of the lag between command and display (96 ms), and the processing code included uses that average lag to correct the target appearance time as much as possible. But without the actual trial-specific timing, a precise correction is not possible. We therefore do not rely on the precise timing of the target appearance in our analyses and we recommend the same to others. The included code which generates the “Processed” format corrects for the reach target appearance timing as much as possible. Please reference this code if you plan to use the “Raw” format as a starting point.

- **cont.vel:** array (#time_bins x 2) containing the x (column 1) and y (column 2) velocities of the arm for each time bin
- **cont.acc:** array (#time_bins x 2) containing the x (column 1) and y (column 2) accelerations of the arm for each time bin. Calculated from the velocity rather than measured directly.

“Processed” format (standard container (Matlab .mat file), custom organization)

- Data from each experimental session is contained in a single .mat file. Such a file contains a *subset* of the behavioral and electrophysiological data for each trial from that session. In particular, it includes a window of this data which flanks the onset of each reaching movement, and which has been processed and down-sampled to 10 ms time bins. This file contains one Matlab structure, **Data**.
 - **Data:** A Matlab structure with the fields given below. Each field contains either a cell array indexed by reach number (#reaches x 1), or an array indexed by trial number (#trials rows, variable column numbers). The fields are:
 - **Data.kinematics:** A cell array (#reaches x 1). Each cell contains an array (#time_bins x 7) with the kinematic data for that reach.
Columns:
 - 1: x position of reach (on-screen cursor, in cm)
 - 2: y position of reach (on-screen cursor, in cm)
 - 3: x velocity of reach (on-screen cursor, in cm/sec)
 - 4: y velocity of reach (on-screen cursor, in cm/sec)
 - 5: x acceleration of reach (on-screen cursor, in cm/sec²)
 - 6: y acceleration of reach (on-screen cursor, in cm/sec²)
 - 7: time stamps for each measurement (in sec).
 - **Data.neural_data_M1:** A cell array (#reaches x 1). Each cell contains an array (#neurons x #time_bins) which contains the spikes for each neuron in M1. Each entry in this array is an integer equal to the number of spikes that occurred for a given neuron in a given time bin.
 - **Data.neural_data_PMd:** Analogous to **Data.neural_data_M1**, but for neurons in PMd.
 - **Data.block_info:** An array (#trials x 23) which contains the experimental data about each trial. This is the same information contained in **trial_table** in the “Raw” format, and the column information can be found there.
 - **Data.trials:** An array (#trials x 1) which denotes which trials were used in constructing this data (i.e., if any of the four possible reaches during a given trial is included in this data). If a trial was used, the entry at the corresponding position in this array is equal to 1. If it was not used, the entry is equal to 0.

- **Data.timestamps:** A cell array (*#reaches x 1*). Each cell contains an array (*#time_bins x 1*) with the timestamp of that bin (in seconds).
- **Data.trial_num:** A cell array (*#reaches x 1*). Each cell contains the trial number from which this reach came.
- **Data.reach_num:** A cell array (*#reaches x 1*). Each cell contains the number of the reach within the trial (i.e., within {1,2,3,4}). Each trial contains up to four reaches.
- **Data.reach_st:** A cell array (*#reaches x 1*). Each cell contains the time that the reach started (in seconds).
- **Data.cue_on:** A cell array (*#reaches x 1*). Each cell contains the *approximate** time that the reach target (go cue) appeared.
- **Data.reach_end:** A cell array (*#reaches x 1*). Each cell contains the time that the reach ended (in seconds).
- **Data.reach_pos_st:** A cell array (*#reaches x 1*). Each cell contains a *1x2* array denoting the starting position of the reach (in cm).
- **Data.reach_pos_end:** A cell array (*#reaches x 1*). Each cell contains a *1x2* array denoting the ending position of the reach (in cm).
- **Data.reach_dir:** A cell array (*#reaches x 1*). Each cell contains the angle of the reach (in radians), based upon the starting and ending positions.
- **Data.reach_len:** A cell array (*#reaches x 1*). Each cell contains the length of the reach (in cm), based upon the starting and ending positions.
- **Data.target_on:** A cell array (*#reaches x 1*). Each cell contains a binary array (*#time_bins x 1*) which indicates *approximately* when the reach target (go cue) appeared (entry = 1 when go cue appeared, entry = 0 otherwise).
- **Data.time_window:** A cell array (*#reaches x 1*). Each cell contains a binary array (*#time_bins_trial x 1*) which indicates which time bins, of all available during each trial, were used for the given reach. There can be up to four reaches per trial, and this array simply indicates the part of the trial that was used.

How to get started

1. Download the data and related Matlab scripts.
2. Open *visualize_data.m* to visualize the data in a number of different ways (neural data, kinematics, etc.). You can also use this to further process the data to generate a data file for analysis (e.g., modeling).

3. Open *clean_data.m* if you'd like to start from the raw data files to generate your own processed files. You can change settings like time bin size, kinematic definition of a reach, etc.

Change History

Version 0.61 (Dec 13, 2018) – Original version.

Version 0.61 (Dec 13, 2018) – Updated reference to Lawlor et. al (2018) to J Comput Neurosci from BioRxiv